1	**(a)**		The principle of conservation of momentum states that total momentum of the system remains constant provided that there is no net/resultant force exerted on the system	 **B1** **B1**
	(b)	**(i)**	Since no net external force, total momentum along AB remains constant. Total initial momentum along AB = Total final momentum along AB $4.0\,(4.2\cos\theta) + 2.8\,(6.0\cos\theta) = (4.0 + 2.8)(3.7)$ $33.6\cos\theta = 25.16$ $\theta = 41.5°$	 **C1** **A1**
		(ii)	Impulse = change in momentum Considering direction perpendicular to line AB Impulse = final momentum – initial momentum $= 0 - 4.0\,(4.2\sin 41.5°)$ $= -\,11.1$ N s (accept positive value)	 **C1** **A1**
		(iii)	Since, the total final momentum along the direction perpendicular to AB (vertical) is zero, the total initial momentum along the direction perpendicular to AB (vertical) must equal to zero. Since both balls have the same initial momentum (16.8 kg m s^{-1}), the angles have to be the same, so that the total initial momentum along the direction perpendicular to AB equals to zero.	**B1** **B1**
		(iv)	Total KE before collision = $\frac{1}{2}(4.0)(4.2)^2 + \frac{1}{2}(2.8)(6)^2$ $= 85.7$ J Total KE after collision = $\frac{1}{2}(4.0 + 2.8)(3.7)^2$ $= 46.5$ J Since the KE is not conserved, the collision is inelastic.	**M1** **A1**
2	**(a)**	**(i)**	1. A gas consists of a very large number of molecules	**B1**